DevOps
Important Commands

- Page 1 – Linux & Git Essentials

Linux Essentials

Powerful commands to navigate, manage and operate your system.

PYTHON LIFE

```
pwd                   # Current directory
ls -la                # List all files (long format)
cd folder             # Change directory
mkdir project         # Create directory
touch file.txt        # Create file
cp source dest        # Copy file or directory
mv old new            # Move or rename
rm -rf folder         # Remove directory (force)
find . -name "*.js"   # Find files by name
grep "text" file      # Search text in file
chmod +x script.sh    # Make file executable
ps -ef                # Show running processes
top                   # Real-time system monitor
df -h                 # Disk space usage
free -m               # Memory usage
```

2 Git Essentials

Track changes, collaborate and manage your code like a pro.

```
git init                 # Initialize a new repository
git clone <repo>         # Clone a remote repository
git status               # Check status of repository
git add .                # Stage all changes
git commit -m "message"  # Commit changes
git push origin main     # Push to remote repo
git pull origin main     # Pull latest changes
git checkout -b feature  # Create & switch branch
git merge feature        # Merge branch into current
git stash                # Save changes temporarily
git log                  # View commit history
```

DevOps Important Commands

- Page 5 – Helm, Monitoring & Networking

Helm

Package Manager for Kubernetes

```
01 helm repo add
02 helm repo update
03 helm install app chart
04 helm upgrade app chart
05 helm list
06 helm uninstall app
```

Monitoring

Observe & Analyze Systems

PYTHON LIFE

```
01 systemctl start prometheus
02 systemctl start grafana-server
03 systemctl status prometheus
04 systemctl status grafana-server
```

Prometheus Metrics Collection

Grafana Visualization & Dashboards

Networking & SSH

Connect, Transfer & Troubleshoot

```
01 ping google.com                          # Check connectivity
02 curl https://example.com                 # Fetch data
03 wget URL                                 # Download file
04 netstat -tulnp                           # Network connections
05 ss -tulnp                                # Socket statistics
06 nslookup google.com                      # DNS lookup
07 dig google.com                           # Detailed DNS lookup
08 ssh user@server                          # Connect to server
09 scp file.txt user@server:/home/          # Secure file copy
10 ssh-keygen                               # Generate SSH key
```

Must-Know Interview Commands

```
01 git status
02 docker ps
03 docker logs
04 kubectl get pods
05 kubectl apply -f
06 terraform plan
07 terraform apply
08 ansible-playbook
09 helm install
10 aws configure
11 curl
12 ssh
```

DevOps Success Path

Git → Docker → Kubernetes → Jenkins → AWS → Terraform

Master Git → Docker → Kubernetes → Jenkins → AWS → Terraform to build a strong DevOps career. 🚀

Page 4

DevOps
Important Commands

- Page 4 – Cloud & Infrastructure as Code

AWS CLI

Cloud Management

Manage and automate AWS services from the command line.

PYTHON LIFE

```
01  aws configure                      # Configure AWS CLI
02  aws s3 ls                          # List S3 buckets
03  aws s3 cp file.txt s3://bucket/    # Copy file to S3
04  aws ec2 describe-instances         # List EC2 instances
05  aws iam list-users                 # List IAM users
06  aws lambda list-functions          # List Lambda functions
```

Terraform

Infrastructure as Code

Provision and manage infrastructure using code, consistently and repeatably.

```
01  terraform init          # Initialize working directory
02  terraform validate      # Validate configuration
03  terraform fmt           # Format code
04  terraform plan          # Preview changes
05  terraform apply         # Apply changes
06  terraform destroy       # Destroy infrastructure
07  terraform output        # Show output values
```

Ansible

Configuration Management

Automate software configuration, deployment and task orchestration.

```
01  ansible all -m ping                # Check connectivity
02  ansible-playbook playbook.yml      # Run playbook
03  ansible-galaxy install role        # Install role from Galaxy
```

Infrastructure Automation Flow

Code (Configurations) → Terraform (Provisions Infra) → Ansible (Configures & Deploys) → Cloud (AWS / Azure / GCP)

DevOps
Important Commands

Page 3 – CI/CD & Build Tools

Jenkins
Automation Server

Automate builds, tests and deployment pipelines with ease.


```
systemctl start jenkins
systemctl stop jenkins
systemctl restart jenkins
systemctl status jenkins
```

Maven
Build Automation

Build, test and manage Java projects with powerful lifecycle management.

```
mvn clean
mvn compile
mvn test
mvn package
mvn install
mvn spring-boot:run
```

Gradle
Flexible Build Tool

Fast, flexible and extensible build automation tool.

```
gradle clean
gradle build
gradle test
gradle run
```

CI/CD Pipeline Flow

Git – Code Commit → Jenkins – Build & Automate → Maven/Gradle – Build & Test → Deployment – Release to Prod

DevOps Important Commands

Page 2 – Docker & Kubernetes

Docker

Docker helps you build, share and run applications in isolated containers.


```
docker images                    # List Docker images
docker ps                        # List running containers
docker ps -a                     # List all containers
docker pull nginx                # Pull nginx image
docker run -d -p 80:80 nginx     # Run nginx container
docker stop container            # Stop container
docker restart container         # Restart container
docker exec -it container bash   # Exec into container
docker logs container            # View container logs
docker build -t app .            # Build Docker image
docker-compose up -d             # Start services in background
docker-compose down              # Stop and remove services
```

Kubernetes

Kubernetes helps you deploy, manage and scale containerized applications.

```
kubectl get nodes                          # List cluster nodes
kubectl get pods                           # List pods
kubectl get deployments                    # List deployments
kubectl get services                       # List services
kubectl describe pod pod-name              # Describe a pod
kubectl logs pod-name                      # View pod logs
kubectl exec -it pod-name -- bash          # Exec into pod
kubectl apply -f deployment.yaml           # Apply configuration
kubectl delete -f deployment.yaml          # Delete configuration
kubectl rollout restart deployment/app     # Restart deployment
```

Pro Tip

Learn Docker before Kubernetes.